

Technical Datasheet — VBF-T10A-DSH-01

Next-generation pure electric sedan cell (virtual design). Values from parameter set / simulation output; sources annotated.

Field	Value	Source
Rated capacity	5.0 Ah nominal / 5.057 Ah (1C DFN verified)	parameter set / `cell/T10_1c_dfn.json`
Nominal voltage / window	2.5 – 4.2 V (midpoint 3.74 V at 1C)	parameter set / `cell/T10_energy_dfn.json`
Rated energy	18.55 Wh (1C discharge integration)	`cell/T10_energy_dfn.json:energy_wh`
Gravimetric energy density	426.89 Wh/kg (contract caliber, electrolyte excluded)	`cell/T10_energy_dfn.json`
Volumetric energy density	899.5 Wh/L (contract caliber)	`cell/T10_energy_dfn.json`
Maximum continuous discharge rate	1C (5 A) verified; 4C charge verified for fast charge	simulation protocol
Fast-charge capability	4C charge @45 °C: T_max 324.55 K (51.4 °C), no lithium plating (anode min +0.0341 V)	`cell/T10_4c_dfn.json`
Operating temperature range	simulated at 298 K (discharge) and 318 K (4C charge); full range Not simulated	simulation protocols
Cycle life	Not simulated (aging model not run for this chemistry) — must not fabricate	—
Safety	no plating at 4C; T_max ≤ 60 °C; overcharge to 4.70 V without thermal runaway (triggered=false)	`cell/T10_4c_dfn.json`, `cell/T10_tr.json`
Dimensions	65 mm × 1580 mm × 200.8 μm (active stack)	parameter set / calc-energy
Mass	43.45 g (active stack, electrolyte & casing excluded)	`cell/T10_energy_dfn.json:mass_kg`
Chemistry	NMC811 / graphite+SiOx (OKane2022)	parameter set